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*Review*

# A Review of On-Field Investigations into the Biomechanics of Concussion in Football and Translation to Head Injury Mitigation Strategies

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**Abstract—**This review paper summarizes the scientific advancements in the field of concussion biomechanics in American football throughout the past five decades. The focus is on-field biomechanical data collection, and the translation of that data to injury metrics and helmet evaluation. On-field data has been collected with video analysis for laboratory reconstructions or wearable head impact sensors. Concussion biomechanics have been studied across all levels of play, from youth to professional, which has allowed for comparison of head impact exposure and injury tolerance between different age groups. In general, head impact exposure and injury tolerance increase with increasing age. Average values for concussive head impact kinematics are lower for youth players in both linear and rotational acceleration. Head impact data from concussive and non-concussive events have been used to develop injury metrics and risk functions for use in protective equipment evaluation. These risk functions have been used to evaluate helmet performance for each level of play, showing substantial differences in the ability of different helmet models to reduce concussion risk. New advances in head impact sensor technology allow for biomechanical measurements in helmeted and non-helmeted sports for a more complete understanding of concussion tolerance in different demographics. These sensors along with advances in finite element modeling will lead to a better understanding of the mechanisms of injury and human tolerance to head impact.

**Keywords—**Head impact, Concussion risk, Linear acceleration, Rotational acceleration, Rotational velocity, Head impact sensor, Helmet.

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## INTRODUCTION

The biomechanics of head impacts and concussion in American football (referred to as "football" throughout this review) have been studied for nearly 60 years. The ability to measure on-field concussion biomechanics has long been a goal. This was first noted by Colonel John Paul Stapp who commented in 1975 that we should instrument football players as soon as miniature sensors become available.[9,126] A number of different factors have motivated this research, including desired improvement in protective equipment,[33,38,146] changes in rules and regulations to increase player safety,[4,60,68] and translation of knowledge to other fields such as motor vehicle and military safety.[21,65] Instrumenting football players on the field provides a method for collecting real-world head impact data in an ethical manner to better understand human tolerance to head impacts.

Advances in sensor technology have made it possible to collect on-field head impact data from football players. Research has focused on all levels of play from youth football to professional. This review will focus on on-field head impact data collection following the timeline of major research efforts in the field (Fig. 1). Early attempts at instrumenting football helmets began in 1961,[108] and provided valuable insight to the challenges associated with on-field data collection. These studies did not result in any widely used technology due to the limitations of systems used. Studies on the biomechanics of head impacts in the National Football League (NFL) began in 1996. Instead of on-field instrumentation, the NFL work involved laboratory reconstructions of head impacts from game videos. Researchers began instrumenting college football players (ages 18–22) in 2003, and high school football



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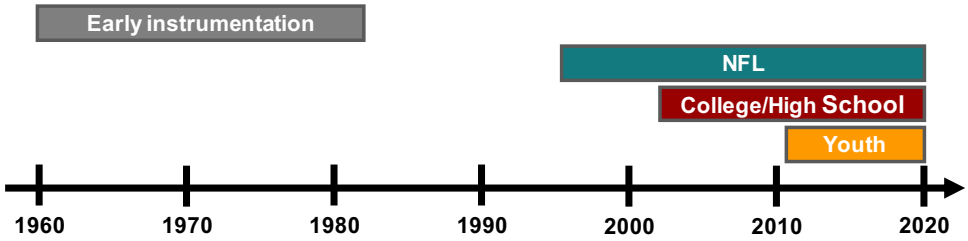


**FIGURE 1. This review will focus on major research efforts for collection of on-field biomechanical head impact data collection. Early attempts at collecting on-field data will be reviewed first, followed by initiatives at each level of play in football: professional (NFL), college and high school, and youth.**

players (ages 14–18) in 2007. The first youth football (under 14 years old) study began in 2011. In this review, a section will be devoted to each major research effort in football concussion biomechanics shown in Fig. 1: early instrumentation, NFL, college and high school, and youth. Within each section, on-field data collection and analysis will be reviewed, followed by translation of that data to injury metrics and risk functions, and helmet evaluation methods that use both risk functions developed from on-field analysis and head impact exposure data. Throughout this review, the term "concussion" refers to a diagnosed concussion in an athlete that has reported symptoms. "Concussive impacts" refer to impacts that result in concussion symptoms and subsequent diagnosis, and "non-concussive impacts" refer to impacts that do not result in any concussion symptoms. Additionally, the term "concussion tolerance" refers to the impact severity (in terms of head kinematics) that an individual or demographic can sustain without showing symptoms of concussion.

It should be noted that although this review focuses on the biomechanics of diagnosed concussions, it is well known that concussions are vastly under-reported in football and other sports.[10,59,63,67,74,150] Athletes are much less likely to report lower-grade concussions either because they do not think it is serious enough or do not recognize it as a concussion.[10,59,67] Additionally, neuroimaging and cognitive changes have been identified after a season of head impacts in football without any identified concussion symptoms.[5,8,17,73,130,133] These findings indicate that repetitive head impacts of lower severity may still have negative neurologic consequences that could lead to neurodegenerative changes later in life.[6,82] There is also evidence to support that increased head impact exposure can predispose athletes to concussion.[12,13,113,128] Therefore, evaluating the biomechanics of diagnosed concussions only provides insight to a subset of injuries, and is likely skewed toward higher-grade injuries as those are more likely to be reported.

## EARLY INSTRUMENTATION

The first attempts to quantify head acceleration in football players were performed in high school and collegiate athletes. Helmet or head-mounted instrumentation was used in several studies between 1961 and 2000.[83,84,88,107,108] Moon *et al.* instrumented high school and college football players with a telemetry system during practices and games.[83] Three accelerometers were located on the top, back, and right side of each player's head within a headband, with the rest of the instrumentation in the shoulder pads connected to the accelerometers with wires. The system was bulky, and instrumentation required 15 minutes of setup prior to each use. Accelerations were only accurate in laboratory tests for impacts that pushed the accelerometers into the headform. Many of the recorded accelerations exceeded 1000 g with no identified injury. Impacts of that magnitude would likely cause a severe head injury given that more recent studies have determined that the average impact resulting in concussion symptoms occurs around 100 g for adult male football players.[12,18,22,103,116,155] Although non-concussive impacts have been reported above 100 g, 1000 g would represent 100% risk of skull fracture.[77] Other early forms of instrumentation used accelerometers attached to the inner surface of the helmet shell or on the suspension straps inside the helmet.[84,107,108] These studies similarly housed the remaining instrumentation within the shoulder pads of players. Small numbers of players were recruited for these studies, which also reported high accelerations with no reported concussion symptoms. Naunheim *et al.* published head impact data from high school hockey, football, and soccer players in 2000.[88] For the football players, a triaxial accelerometer was embedded in the foam padding of a football helmet. The acceleration magnitudes were much lower than previous studies, with an average of 29 g and range of 10 to 120 g.

## NFL

### *On-Field and Laboratory Data Collection*

The first series of experimental studies focused on concussion in professional American football was published between 2003 and 2007.[96–104,140–142,144,145] These studies were funded by the National Football League (NFL), and evaluated clinical, biomechanical, and epidemiological factors associated with concussion. The biomechanics of concussive and non-concussive impacts were determined through reconstructions of impacts that were video recorded during NFL games.[90,91,103] Both player-to-player and player-to-ground impacts were reconstructed in the laboratory using Hybrid III anthropomorphic test devices (ATDs). Thirty-one impacts were reconstructed, which included 25 concussed players and 33 not concussed. The average linear and rotational accelerations for concussive impacts were 98 ± 28 g and 6432 ± 1813 rad/s$^2$. Average non-concussive impacts had accelerations of 57 ± 22 g and 4028 ± 1438 rad/s$^2$. The NFL reconstructions and associated video analysis of impacts have been used to further understanding of brain injury mechanisms through computational modeling.[142] The reconstructions were modeled to determine strain distributions in the brain throughout impact, which showed a lag between peak strain values and peak head acceleration. Functional deficits also correlated with regional brain strain.

These reconstructions have several limitations, including error in impact velocity and location determined from video analysis, biofidelity of the ATDs used in reconstructions, and sampling of concussive and non-concussive impacts. The sampling of impacts was heavily skewed toward injurious impacts, with approximately equal numbers of injured and non-injured players. Since sampling was not representative of concussion incidence, the risk curves developed from this data overestimate risk for linear and rotational head kinematics.[115] Additionally, the initial NFL reconstruction data were re-analyzed in 2019 due to inconsistencies noted in the original instrumentation used.[122] Consistency checks could be performed because a nine-accelerometer array was used to determine head kinematics in the original reconstructions, which has redundant linear acceleration measurements. Multiple reconstructions had inconsistent sensor arrays, which resulted in decreased median peak rotational velocity of the concussed players. Because brain strain is heavily dependent on rotational velocity values during impact, finite element analysis of the inconsistent data resulted in average error of 20% in maximum principle strain. Linear head kinematics were not affected by sensor inconsistencies, and rotational acceleration was minimally affected. Despite the limitations associated with NFL reconstructions, the study provided some of the first reasonable estimates of head impact kinematics associated with concussion in professional football players. Even with drastically different methods, similar peak head acceleration values were associated with concussion in collegiate football players.[116]

More recent reconstructions of NFL impacts were performed as part of a larger effort to determine repetitive head impact profiles for different player positions.[56] Exemplary impact types from video analysis of NFL games were reconstructed with a pneumatic ram or drop tower for shoulder-to-head, hip/thigh-to-head, head-to-head, and head-to-ground impacts. The ATD head kinematics recorded during reconstructions were used as input to a finite element brain model to determine brain tissue strain during these types of impacts. Through analyzing impact frequency and time between impacts from game video, and brain tissue strain from coupled physical reconstructions and finite element modeling, head impact profiles were identified for different player positions. Quarterbacks, wide receivers, and defensive backs had the lowest frequency of impacts but greatest brain strain magnitudes, while offensive and defensive linemen had the highest frequency of impacts paired with lowest strain magnitudes.

Several recent studies compared laboratory grade head kinematic measurements to video-based measurements for NFL-type head impacts. The first evaluated the impact of frame rate on determining helmet kinematics since all NFL games are recorded, but some are recorded at a lower frame rate.[7] Generalized NFL head impact scenarios were staged on a football field using Hybrid III ATDs that either impacted another ATD or the ground. The high frame rate (240 Hz) was sufficient for determining changes in helmet translational and rotational velocity when compared to motion tracking, while the low frame rate (60 Hz) could be used to determine pre-impact translational velocity of the helmet but not impact kinematics. Another study compared relative motion between the head and helmet using a similar experimental setup.[55] Peak resultant linear acceleration of the helmet was 2–5 times higher than the head, indicating that helmet kinematics are not indicative of head kinematics. Similar findings have been reported previously for lower magnitude head impacts.[72] These studies indicate that video methods cannot accurately determine head kinematics since only helmet motion can be tracked, but still have use for determining pre-impact velocity as there is minimal relative motion between the head and helmet prior to impact.

*Injury Metrics and Concussion Risk*

Injury metrics are used in safety standards and relative performance evaluation in automotive and sports equipment safety assessments.[85,86] Laboratory impact tests or finite element model simulations are used to generate metrics that can be compared to a minimum safety threshold or relative performance between products. Head injury metrics have been used to reduce different types of injuries through automotive and helmet evaluations. For example, the National Operating Committee on Standards for Athletic Equipment (NOCSAE) has greatly reduced the number of severe and fatal head injuries in football through a helmet certification program that requires all helmets to produce a severity index (SI) below 1200 for impact tests.[78,86] Relative performance ratings for football helmets have evaluated the ability of helmets to reduce concussion risk.[115] Injury metrics can be either a single kinematic variable, a function of kinematic variables, or values derived from finite element simulations. Risk analyses can be performed to determine the likelihood of injury for defined impact conditions.[45–47,78,103,115–117]

The first injury metric based on NFL reconstruction data was head impact power (HIP).[92] HIP was based on the hypothesis that the threshold for head injury would be a critical value of kinetic energy of the head. Since approximations of the Wayne State Tolerance Curve (WSTC) could be rewritten in a form that is proportional to the rate of change in kinetic energy of the head, or power, an expression representing power was chosen for the criterion:

$$\mathrm{HIP} = 4.5a_x\int a_x \mathrm{d}t + 4.5a_y\int a_y \mathrm{d}t + 4.5a_z\int a_z \mathrm{d}t +$$

$$0.016\alpha_x\int \alpha_x \mathrm{d}t + 0.024\alpha_y\int \alpha_y \mathrm{d}t + 0.022\alpha_z\int \alpha_z \mathrm{d}t$$

With the coefficients equal to the mass and moment of inertia of the human head. The coefficients could be varied based on directional sensitivity of the head to impact. HIP risk curves were developed from NFL reconstructions of concussive impacts in professional football players. For these cases, HIP was a better predictor of mild traumatic brain injury (mTBI) than HIC.

Concussion risk curves were generated from the NFL reconstruction data described above for the following metrics: head injury criterion (HIC), severity index (SI), linear acceleration, rotational acceleration, peak linear velocity change, and peak rotational velocity.[103] Logist analysis was used to generate the curves using concussive and non-concussive data points. The risk curves were generated for the struck players alone, and for a combination of striking and struck players in player-to-player collisions. The functions based on struck players alone were thought to be more valid because the loading conditions were similar in those cases. Linear and rotational acceleration values for 50% risk were estimated to be 52 g and 3349 rad/s$^2$ for analysis of the struck players alone, and 81 g and 5488 rad/s$^2$ for the combination of struck and striking players.

More recently, a new metric was proposed for use in evaluating NFL football helmets that combines rotational acceleration and velocity, and HIC, which is a function of linear acceleration.[44] The metric is termed combined metric (CM):

$$\mathrm{CM} = \frac{1}{24}\sum\left(\frac{\alpha_{\mathrm{peak}}}{\alpha_{\mathrm{ave}}} + \frac{\omega_{\mathrm{peak}}}{\omega_{\mathrm{ave}}} + \frac{\mathrm{HIC15}}{\mathrm{HIC15}_{\mathrm{ave}}}\right)$$

where peak rotational acceleration ($\alpha_{\mathrm{peak}}$), peak rotational velocity ($\omega_{\mathrm{peak}}$), and HIC for each impact are normalized with the average values for all other helmet models tested each year. Lower combined metric values indicated better performance. Because the combined metric is based on average kinematics of all helmet models tested, it represents a relative metric for comparing impact performance between helmets. The values cannot be used to determine likelihood of head injury.

The NFL reconstructions have also been used to develop injury predictors through the use of a finite element model.[62] Eight different tissue injury predictors were compared with 10 kinematic-based predictors to determine the highest correlations with injury. A linear combination of peak change in rotational velocity and HIC had a high correlation with strain and was suggested as a potential injury predictor, although several other combinations had similarly high correlations with injury.

$$\mathrm{KLC} = 0.004718\Delta\omega_r + 0.000224\mathrm{HIC}$$

This metric has been referred to as the Kleiven criterion or Kleiven's linear combination (KLC).

Laboratory impacts based on NFL impact conditions have been used to develop several injury metrics in conjunction with finite element modeling. Two recently developed metrics are the universal brain injury criterion (UBrIC), and Diffuse Axonal Multi-Axis General Evaluation (DAMAGE).[48,49] Both criteria are based on mechanical systems analysis, estimating strain within brain tissue from input head kinematics. UBrIC assumes that brain deformation is analogous to deformation of a second-order mechanical system, and uses critical values from FE simulations to predict strain-based metrics from input rotational velocity and acceleration. The critical values were determined from a wide range of impact reconstructions, including NFL impact conditions determined from video analysis.[146]

DAMAGE is based on equations of motion of a 3DOF coupled second order system. The metric predicts maximum strain in the brain from rotational acceleration time history. The same NFL-level impacts were used for performance assessment of the metric. The advantage of these metrics is that they provide rapid estimation of brain tissue strain from head impact kinematics rather than through finite element simulations which can be computationally expensive and time consuming.

### *Helmet Evaluation*

Relative impact performance ratings of helmets complement existing standards that all football helmets are required to pass to be sold in the US.[86,87] Helmet standards have drastically reduced the number of deaths and severe head injuries in football.[78] Several testing protocols have been used to evaluate differences in NFL helmet performance using on-field data from video analysis of games and laboratory reconstructions.[104,143,145,146] In general, these studies found that helmet performance differs between models in terms of acceleration magnitudes and risk of concussion. Risk of concussion was determined through logistic regression analysis of kinematic variables from reconstructed NFL head impact cases. Helmet models have also improved over time in terms of their ability to attenuate head kinematics during impact.[104,143,145,146] A uniform method for quantifying relative performance of NFL helmets was introduced in 2017, based on a previously proposed method by Viano *et al.*[44,146] The method uses a pneumatic ram with a nylon spherical cap over a layer of vinyl nitrile foam (Fig. 2). The ram impacts a Hybrid III head and neck assembly attached to a sliding mass that moves in the direction of impact. Impact conditions (locations and velocities) were selected based on video analysis of concussions and severe impacts in NFL games.[102,103] The eight impact locations include four on the helmet shell, two central on the facemask, and two at the edge of the facemask. Each location is impacted at three velocities: 5.5, 7.4, and 9.3 m/s. A new performance metric, combined metric, was developed to rank helmet models relative to one another.

Hybrid III head-neck assembly and impactor finite element models were developed to evaluate football helmet design modifications.[50] These models are open source, and are intended for manufacturers to use for optimizing helmet design prior to production of new models. The models were validated by bare-head laboratory tests used by the NFL and other research groups.[32,44,120,146]

Recent video analysis of concussion in the NFL showed a higher proportion of helmet-to-body and helmet-to-ground impacts than previously reported.[69] The biomechanics of helmet-to-ground impacts were specifically analyzed due to the increase in frequency and differing impact conditions.[58] The results can be used to inform injury mitigation measures and helmet testing methods.

## COLLEGE AND HIGH SCHOOL

### *On-Field Data Collection*

The first widely used helmet-mounted instrumentation for measuring head impact kinematics is the head impact telemetry (HIT) system. The HIT system has been used to collect a large dataset of head impacts from college, high school, and youth football teams, and has been extensively validated.[11,34,54,72,125] It is a commercially available system that measures resultant linear acceleration and estimates resultant rotational acceleration from six single axis accelerometers that are spring mounted to stay in contact with the head.[34] The instrumentation fits inside the shell of the helmet between existing padding. Since the accelerometers stay in contact with the head, they measure head acceleration rather than helmet acceleration, which is substantially higher.[72] A six-degree-of-freedom system was also developed for a better understanding of head kinematics during impact, but that system is not widely used.[114]

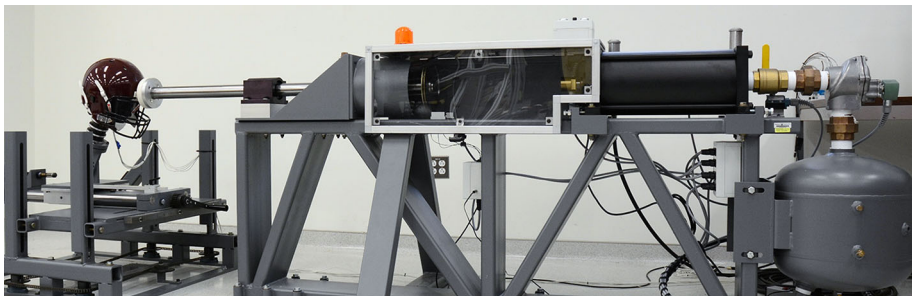

**FIGURE 2. Pneumatic ram used in NFL helmet evaluation protocol, manufactured by Biokinetics, Canada. The ram impacts a helmeted headform attached to a sliding mass at eight impact locations and 3 impact speeds.**

The HIT system was first used in collegiate football players in 2003, and has since been used to collect a large dataset of concussive and non-concussive impacts (Fig. 3).[42] These data have been used to extensively describe head impact exposure in football players at different levels of play. A number of studies have shown that head impact exposure in terms of frequency and magnitude varies by player position,[19,20,27,36,37,68,79,123] session type,[20,37,64,79,139] and impact location.[13,35–37,64,79,112,115] In general, head impacts tend to be more frequent during games than practices, and skill positions tend to have higher magnitude impacts while linemen have a higher frequency of head impacts. For location, impacts to the front of the helmet are most common, and impacts to the top of the head generally have higher linear acceleration and lower rotational acceleration. Additionally, head impact exposure varies by level of play, with collegiate players sustaining high magnitude impacts more frequently than high school players.[123]

In 2014, the the National Collegiate Athletic Association (NCAA) and U.S. Department of Defense (DoD) partnered to form the Concussion Assessment, Research, and Education (CARE) Consortium. It is the largest prospective concussion study to date, and has enrolled over 41,000 male and female athletes. The CARE Consortium combines biomechanical and clinical concussion data to better understand the neurobiological natural history of concussion and to determine robust markers of injury and recovery. The findings of this research will help inform policy decisions concerning concussion in athletes and military service members. Six institutions have been collecting head impact data from athletes for the last 5 years, helping to create the largest dataset of college football head impacts. The universities and military academies collecting biomechanical data are: the U.S. Air Force Academy, the U.S. Military Academy at West Point, the University of California Los Angeles, the University of North Carolina at Chapel Hill, the University of Wisconsin, and Virginia Tech. A special issue was published in the Annals of Biomedical Engineering in 2019 focused on NCAA-DoD CARE Consortium research.[3,27,30,53,110,113,121,127,128,147]

**FIGURE 3. The HIT system was first used at Virginia Tech in 2003. Helmet instrumentation wirelessly transmits impact data to a signal receiver and laptop on the sidelines (pictured here).**

Data collected through the CARE Consortium was used to quantify specific factors that affect head impact exposure in college football players. Exposure varied by team, player position, and player ability, but individual differences still accounted for half of the variance in exposure when controlling for those factors.[27] This study supports the need for considering head impact exposure and concussion risk on an individual basis. Another study analyzed head impact exposure for specific practice drills, and indicated practice structure changes that could reduce avoidable head impacts from teammates.[3] One practice structure change aimed at reducing head impact burden for college football players was elimination of two-a-day practices during preseason in 2017.[127] However, the number of preseason contact days increased, so the overall preseason head impact exposure increased. This paper highlights the importance of making data-driven decisions when aiming to reduce injury risk to players.

The large number of head impacts collected with the HIT system has also facilitated quantification of head kinematics of concussive impacts. For high school and collegiate athletes the average linear and rotational accelerations for concussive impacts are 104 ± 30 g and 4726 ± 1931 rad/s$^2$.[116] Although these averages give an idea of the magnitude of concussive impacts, no association between impact magnitude and acute clinical outcome has been observed.[52] There does appear to be an association with the timing of concussion diagnosis and impact magnitude, with immediately diagnosed concussions being higher magnitude.[13] There is also some evidence that the impacts leading up to the concussive impact may predispose some athletes to injury. It has been shown that concussed players have greater head impact exposure on days of injury than other days, and greater head impact exposure than matched controls.[12,128] The large amount of variation in concussive impact magnitudes means that many concussive impacts do not stand out among impacts that did not cause a concussion. However, if the impacts are evaluated on an individual basis, concussive impacts are among the most severe for that individual.[119] Additionally, if concussed athletes are compared to physically matched controls (similar

height, weight, age, race, concussion history), they have a greater frequency of impacts overall, greater frequency of high magnitude impacts, and greater risk-weighted exposure.[113] These studies support the need to determine which traits have the most influence on concussion tolerance, so that risk analyses can be performed on a more individualized basis according to those traits.

Although the HIT system has provided valuable data and led to a better understanding of the biomechanics of concussion, there are limitations to the system. The commercially available version can only be used in football, preventing analysis of sex differences in concussion tolerance. Since the sensor array is helmet mounted, there is some relative motion between the helmet and head that make the measurements less accurate. Although the average error in aggregate data from a population is low, individual data points are less reliable and should be used with caution.[11,54] Due to these limitations, researchers have investigated alternative types of sensor mounting for more accurate and universal use. Alternative options that have been proposed include headband- or skullcap-mounted, skin-mounted, earpiece, and mouthguard sensors.[9,23,24,66,95,125]

The skin-mounted sensor xPatch has been widely used,[8,61,68,109,124,130,152] including for an evaluation of a helmetless-tackling intervention to reduce head impact exposure in collegiate football players.[131] Although the skin-mounted sensors have performed well in laboratory validation tests, on-field and other human subject evaluations have shown that both impact frequency and magnitude are overpredicted in real-world conditions.[106,136,151] Relative motion between the skin and skull during impact leads to overprediction of acceleration values and false positive impacts from normal motions like jumping.

Mouthguard sensors are more tightly coupled to the skull and therefore have potential to more accurately measure head impact kinematics on the field.[151] Mouthguard sensors also have the advantage of use in helmeted and non-helmeted sports. Current mouthguard sensors include one originally developed by X2 Biosystems that has been modified by a team at Stanford, the Impact Monitor Mouthguard by Prevent Biometrics, and the Vector MouthGuard by i1 Biometrics, and one developed by researchers at Wake Forest University.[9,24,76,110] These mouthguards typically are highly correlated with laboratory-grade sensors in ATD impact tests, and show potential for more accurate measurements of on-field head kinematics. Due to the wide variability of commercially available sensor performance, a laboratory methodology was developed that combines helmeted and unhelmeted ATD imacts to objectively measure sensor performance by comparing the results to reference sensors.[136] Sensor technology is rapidly changing, and as systems improve, ones that are more tightly coupled to the skull such as custom mouthguards or mouthpieces will provide more accurate measurement of head kinematics.

### *Injury Metrics and Concussion Risk*

Several studies quantifying concussion biomechanics at the high school and college levels evaluated the predictive capability of different injury metrics.[22,51] For a study combining collegiate and high school head impact data, principal component analysis with impact location weighting was used to develop an injury metric that combined SI, HIC, linear acceleration, and rotational acceleration.[51] The predictive capability for concussion of this new metric, now known as HITsp, was compared to linear acceleration, rotational acceleration, and HIC. The combined metric was more predictive than any of the individual variables. A study of concussions in high school football found that a combination of rotational acceleration, linear acceleration, and impact location was the best predictor of injury when using a classification and regression tree analysis.[22]

Risk analyses have also been conducted based on on-field concussion data collection. Concussion risk for college football players was first estimated from HIT system data with only 4 concussions in 27,000 head impacts recorded.[46] Risk functions were generated for linear acceleration and HIC values. The NFL risk functions predicted much higher risk of concussion than this risk function for the same impact severity because of the incidence of concussion in each dataset.[103] The HIT system dataset more accurately represented injury incidence since it was a sample of real-world data, while the NFL dataset was heavily weighted toward concussive impacts. Since a large number of non-concussive impacts existed for the same magnitude as the concussive impacts, risk of injury was much lower than predicted with the NFL dataset. As more concussive impacts were recorded with the HIT system, new risk functions were developed based on linear acceleration, rotational acceleration, and a combination of both.[115–117]

On-field head impact data collected from high school and collegiate football players have been used in development of finite element brain injury predictors or metrics. In one study, previously recorded head impacts from high school and college athletes were simulated with two different finite element head models.[14] Impacts with immediately diagnosed concussion along with control impacts that did not result in concussion from the same players were simulated.

Intracranial response measures of stress, strain, and pressure were compared to kinematic variables for sensitivity and specificity for diagnosing concussion. For this dataset, the intracranial measures were not more sensitive to diagnosed concussion than kinematic measures alone.

An injury metric based on rotational head kinematics was developed through finite element modeling, known as brain injury criterion (BrIC).[132] Scaled animal head injury data were used to determine appropriate thresholds for measures of strain within the brain for predicting concussion. A combination of laboratory ATD tests and on-field football head impact data were used to determine appropriate critical values.

### *Helmet Evaluation*

The risk functions generated from HIT system data were used to evaluate helmet performance. The first football helmet rating methods were known as the Summation of Tests for the Analysis of Risk (STAR) evaluation system.[115] The STAR system involves a series of laboratory impacts to a helmeted headform, and is based on two fundamental principles: that tests are weighted based on how frequently similar impacts occur on the field, and helmets that lower head acceleration will reduce the player's risk of concussion. Drop tests were initially used for football helmet ratings to align with helmet certification protocols used at that time. The STAR equation is used as a metric to summarize all laboratory impacts, and to rank relative performance of helmets:

$$\mathrm{STAR} = \sum_{L=1}^{4} \sum_{H=1}^{6} E(L,H) \cdot R(a)$$

where $L$ represents impact location, $H$ represents drop height (or impact velocity), $E$ is head impact exposure as a function of location and impact velocity, and $R$ is concussion risk as a function of peak resultant linear head acceleration ($a$). The STAR value represents predicted injury incidence for a player wearing the helmet tested for one season if they were subjected to the impact conditions represented by the testing methods. The locations and impact velocities used were based on generalized impact conditions recorded with the HIT system. Exposure was also based on HIT system data, with the weighting determined from how frequently similar impacts occurred on the field. Concussion risk as a function of linear acceleration alone was used since the headform was rigidly attached to the drop rig and did not allow for rotation of the head during impact. Large differences in helmet performance were quantified, providing consumers with guidance on which helmets are safer, and manufacturers with an objective measure of performance to improve helmet designs. Agreement was seen between real-world concussion rate differences and predicted differences by the STAR methodology for 2 of the helmet models evaluated, indicating that the tests were capable of quantifying differences in the ability of helmets to reduce concussion risk.[118]

The football helmet rating protocol has been updated since it was initiated.[138] In the current methods, a pendulum impactor strikes a helmeted head and neck assembly attached to a sliding mass that allows for motion in the direction of impact (Fig. 4). In the original helmet testing protocol, the headform was rigidly attached to a drop carriage.[115] The rigid attachment did not allow the head to rotate, so only linear acceleration could be evaluated for those tests. Since a neck is included in the current test setup, rotational acceleration can be evaluated. The pendulum was chosen for increased repeatability and reproducibility compared to other impact systems.[104] Concussion risk is calculated from a combination of linear and rotational acceleration.[116] The general methods used to evaluate football helmets have also been adapted to youth football helmets, hockey helmets, bicycle helmets, soccer headgear, and flag football headgear.[15,16,25,115,120,137,138]

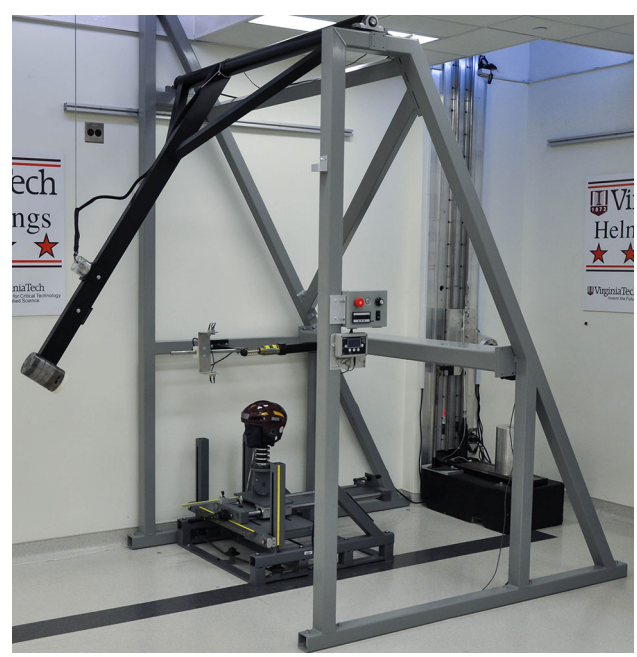

**FIGURE 4. A pendulum impactor is currently used for adult football helmet ratings developed at Virginia Tech. The pendulum impacts a helmeted headform attached to a sliding mass at four locations and three velocities.**

## YOUTH

### *On-Field Data Collection*

Researchers began on-field biomechanical data collection of youth football head impacts in 2011 (Fig. 5).[39] The first few papers published on this data focused on quantifying head impact exposure for players under 14 years old using the HIT system. A progression was seen for the number of impacts per player per season, with older players experiencing more impacts. On average, 7–8 year olds experienced 107 impacts per season, 9–12 year olds experienced 240 impacts per season, and 12–14 year olds experienced 275 impacts per season.[31,39,40] Even within those age groups, the more experienced players tended to have higher head impact frequencies.[57,153] The number of impacts per season were also substantially lower than those seen in high school and college football players. Similarly, head impact magnitude increased with player age, but even the youngest players still had high magnitude impacts that could represent substantial risk of concussion for those players. A portion of the youth football head impact exposure data collected was used to evaluate brain strain using an anatomically correct brain finite element model.[57,81] The impacts were generalized to 6 locations of interest, and the 5th, 50th, and 95th percentile impacts were simulated at each location. The results showed a directional dependence for peak strains within the brain, with locations that resulted in a higher relative amount of rotation resulting in greater strain values.

One trend that has been noticed with younger youth football players is that more high magnitude impacts occur during practices than games.[39,57] These findings indicate that changes in practice structure have potential to greatly reduce head impact exposure in young athletes. Given the evidence that repetitive head impact exposure even without a diagnosed concussion may contribute to neurodegenerative processes later in life, limiting unnecessary head impacts should be considered a priority.[5,41,82] Following publication of the first exposure paper in 2012, the youth football organization Pop Warner instituted rule changes that limited contact during practices and eliminated full-speed head-on drills that resulted in the highest magnitude head impacts.[39] Additionally, USA Football (a governing body of youth football) initiated the Heads Up Football program in 2012, which involved coach education on player safety issues like concussion, and proper equipment fitting and tackling techniques. The combination of limiting contact during practices, educating coaches and players, and using safety equipment properly resulted in reduced head impact exposure and fewer concussions overall.[31,60] Evidence of greater head impact exposure during contact drills and scrimmages motivated a more detailed evaluation of head impact exposure for different drill types.[28,154] Analysis of exposure by practice drill type showed that tackling drills have a greater proportion of high magnitude impacts. A suggested practice structure modification showed that if the teams reduced the time spent on tackling or blocking drills by 10 minutes per practice, high magnitude impacts could be reduced by 38%.

### *Injury Metric and Concussion Risk*

It is likely that age-related differences in concussion tolerance exist for a number of reasons. From a purely biomechanical perspective, a smaller pediatric brain would be expected to have a greater tolerance to head acceleration because a larger amount of rotational acceleration would be needed to produce the same amount of strain as an adult brain.[93,94] However, that analysis assumes that pediatric brains are smaller versions of adult brains. Material testing and modeling have shown that age-related changes in brain material properties influence the mechanical response of the brain to rotational loading.[105,134] Some have also

**FIGURE 5. The first youth football team instrumented with the HIT system was the Auburn Eagles (Riner, VA) in 2011. The instrumented players were 7–8 years old, and the on-field data collection provided the first description of head impact exposure in youth football players.**

suggested that physiological and structural differences in the developing brain may make it more vulnerable to concussion.[2,29] To address these gaps in knowledge, a youth-specific concussion risk function was developed.[25] The analyses were performed with on-field head impact data collected from youth football teams between 2015 and 2018, with a total of 15 diagnosed concussions. Players between the ages of 9 and 14 were included as they participate in tackle football below the high school level. The risk function uses an adaption of a previously developed injury metric for traumatic brain injury, the Generalized Acceleration Model for Brain Injury Threshold (GAMBIT).[89,90] GAMBIT is a function of both linear and rotational acceleration normalized with critical values, and is thought to be analogous to combined axial and shear stresses in engineering analysis. Since the original metric was proposed for more severe brain injury than concussion, the critical values were updated for youth-specific concussion risk. Survival analysis was then used to generate the injury risk curve, with the adaption of GAMBIT used as a predictor of concussion. Based on the risk function and average concussion accelerations observed, youth athletes exhibited a lower tolerance for concussion.

### *Helmet Evaluation*

The youth-specific concussion risk function was used in the development of a helmet performance evaluation protocol.[26] The methods are an adaption of the adult football helmet evaluation system (STAR) mentioned above.[115] For laboratory tests, a youth-specific surrogate was developed, which consisted of a head and neck assembly, and a sliding mass that simulates effective mass of the torso. The surrogate was modeled after a 50th percentile 10–12 year old male. A pendulum impactor is used to strike the surrogate at four locations and three impact velocities, resulting in 12 unique impact conditions. Risk of concussion is determined for each impact test, and weighted based on analysis of on-field youth football head impact exposure by impact location and velocity. The results of all impact conditions are summed for a single metric that can be used to compare relative performance of youth football helmets. A wide range of head accelerations are seen for different helmet models for the same impact conditions (Fig. 6). Linear and rotational acceleration ranged from around 60 g and 2000 rad/s$^2$ to 135 g and 6000 rad/s$^2$ for the front location, high velocity (4.9 m/s) impact. Consumers can use the publicly available helmet ratings to inform purchasing decisions, and manufacturers can use the methods to optimize helmet design.

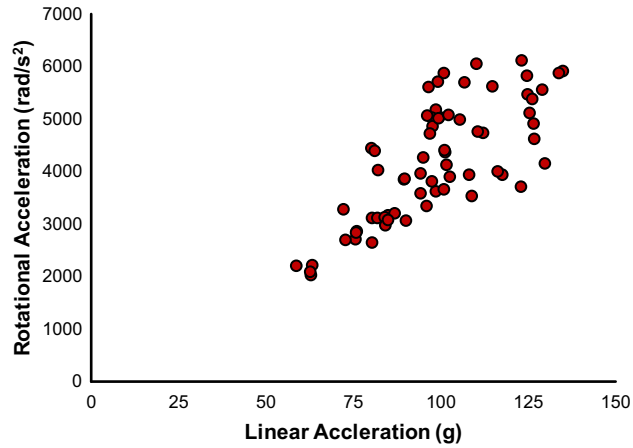


**FIGURE 6. Linear and rotational acceleration for all youth football helmet impacts for the front location, 4.9 m/s velocity condition. Helmet performance varies widely for different helmet models, with linear and rotational acceleration ranging from around 60 g and 2000 rad/s$^2$ to 135 g and 6000 rad/s$^2$ for the same impact.**

## DISCUSSION

On-field head impact data collection across different levels of play has allowed for comparisons between different age groups in terms of head impact exposure and concussion tolerance. Both impact frequency and magnitude increase with player age.[31,40,57] These differences can be partially attributed to increased numbers of practices and games as players advance, but is also likely due to the increased size and aggressiveness of older, more skilled players. Although researchers have expected that concussion tolerance changes with age, the on-field data from youth football players has provided the first *in-vivo* evidence to support that hypothesis.[2,25,29,93,94,105,134] Based on the data collected so far, it appears that pediatric concussions occur at a lower acceleration magnitude than adult concussions, indicating a lower tolerance in the youth population (Table 1). This is the opposite trend that would be expected if pediatric brains were scaled down versions of adult brains with the same material properties and physiology, indicating that the material properties of the brain and/or physiology play a greater role in determining tolerance in the developing brain.[93,94] There are likely changes that occur within the age groups presented here (NFL: 20 years and older; College and high school: 14-22 years old; youth: under 14 years old), with more rapid changes at the youth level.[71] Other factors that may contribute to apparent differences in tolerance are the player's willingness to report injury, and selection for greater brain injury tolerance at more advanced levels of play in football.[111] Players that are more susceptible to con-

**TABLE 1. Comparison of average concussion magnitudes across all levels of play in football.**

| Level of play | Average linear acceleration (g) | Average rotational acceleration (rad/s$^2$) |
|---|---|---|
| NFL[104] | 98 ± 28 | 6432 ± 1813 |
| College/high school[118] | 104 ± 30 | 4726 ± 1931 |
| Youth[29] | 62 ± 30 | 2609 ± 1591 |

cussion are less likely to continue to play a high-impact sport like football long-term.

For the on-field and reconstruction data presented in this review, NFL and college/high school players had similar average concussion magnitudes. Linear accelerations matched very closely with 98 and 104 g for NFL and college/high school players respectively. Rotational accelerations were more variable, with an average of 6432 rad/s$^2$ for NFL players and 4726 rad/s$^2$ for college and high school players. The differences may be due to differences in data collection methods, since the NFL concussions were reconstructed from video analysis, and the college and high school data was from helmet-mounted instrumentation.

While the average concussive acceleration magnitudes are representative of injury distributions, they do not account for the distribution of impacts that did not result in injury. There is overlap between concussive and non-concussive acceleration magnitudes, which determines the shape of the risk curve generated from a given dataset. The risk curves generated from NFL reconstructions had approximately equal numbers of concussive and non-concussive data points, which is not representative of on-field data given estimations that for every concussive impact there are 1600 non-concussive impacts in the NFL.[103,115] Without accounting for the incidence of concussion, risk is likely to be overestimated. The risk curves generated from on-field head impact data are more representative of actual concussion incidences since they include a large number of non-injurious impacts.

On-field head impact data from football players has advanced our understanding of concussion biomechanics. However, there are still many gaps in knowledge that exist. The following knowledge gaps have been identified through this review, and should be prioritized for future research initiatives:

- *Concussion tolerance as a function of age*. Continued data collection at the youth level in football and other sports will lead to a better understanding of how concussion tolerance changes with age. Improvements in head impact sensors will also provide more reliable measurements when smaller sample sizes are available.
- *Sex differences in concussion tolerance*. Further on-field investigations in other sports will lead to a better understanding of sex differences in concussion tolerance. While studies have been performed in other sports evaluating female head impact exposure and concussion biomechanics,[43,70,75,80,129,135,148,149] concussion data are limited, and the sensors used are subject to the limitations described above.[1,106,136,151] Recent advances in wearable sensors will help address many of these knowledge gaps, as well as provide more accurate measurements of head impact biomechanics.
- *Individual concussion tolerance*. As the dataset of head impact kinematics expands and extends to other sports, further analysis of potential factors that increase concussion risk (physical traits, medical history, *etc.*) will provide insight to more individualized treatment of concussion risk.[27,113,119]
- *Effects of head impact exposure on concussion tolerance and long-term neurodegenerative outcomes*. With growing evidence that repetitive head impacts can not only influence immediate concussion tolerance,[12,13,113,128] but also cause neuroimaging and cognitive changes without concussion symptoms that may lead to neurodegenerative processes later in life,[5,6,8,17,73,82,130,133] it will be important to determine best practices for reducing head impact exposure and minimizing adverse outcomes.

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